



MCA504

Programming Logic and Techniques Using Python

Introduction to Python

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Programming in Python



- Python is a general purpose programming language which is:
 - Open source
 - Object-oriented
 - Interpreted
 - high-level programming language with dynamic semantics
- Used by hundreds of thousands of developers around the world in areas such as:
 - Internet scripting
 - System programming
 - User interfaces

Evolution of Python

- Python was created by Guido Van Rossum, a Dutch programmer, and released on February 20, 1991.
- Van Rossum created Python to be a general-purpose tool that could be applied to anything, and to reduce the complexity of coding syntax.

“Python is an experiment in how much freedom programmers need. Too much freedom and nobody can read another's code; too little and expressive-ness is endangered.”



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Did You Know???

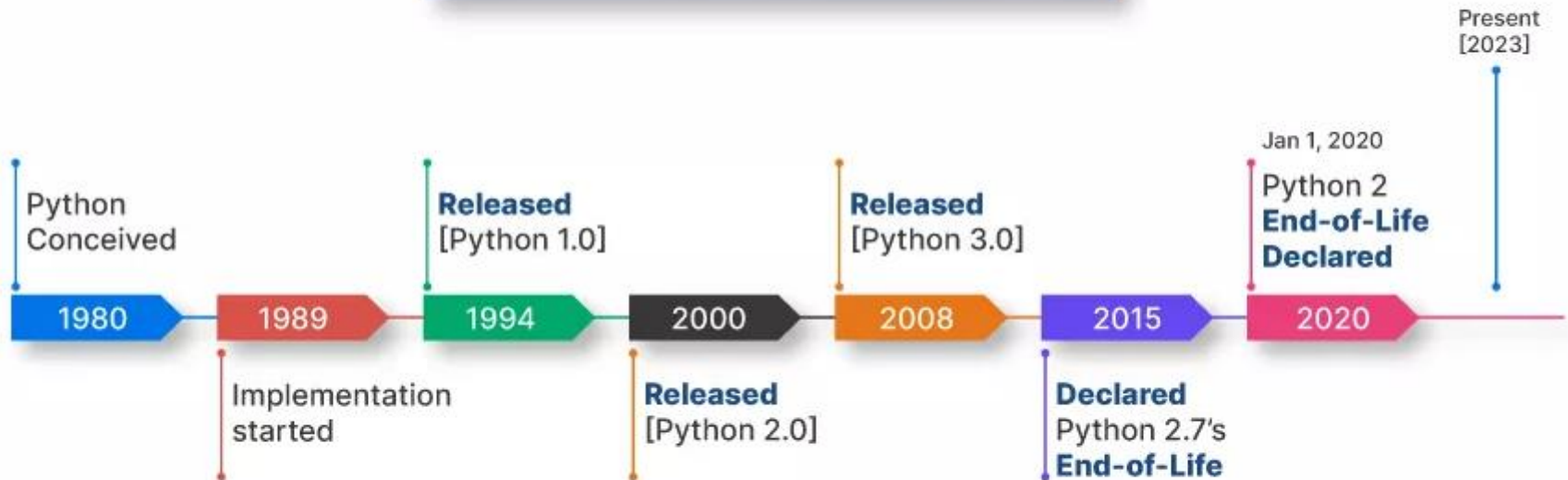
Python was named after an old BBC television comedy sketch series called Monty Python's Flying Circus.

- After releasing the program, Van Rossum further outlined his vision for Python:
 - To be an easy and intuitive language that is just as powerful as those of the major competitors
 - The ability for anyone to contribute to its development by being open source
 - For the language to be easily understandable as plain English
 - That the language be suitable for everyday tasks and allow for short development time

Evolution of Python

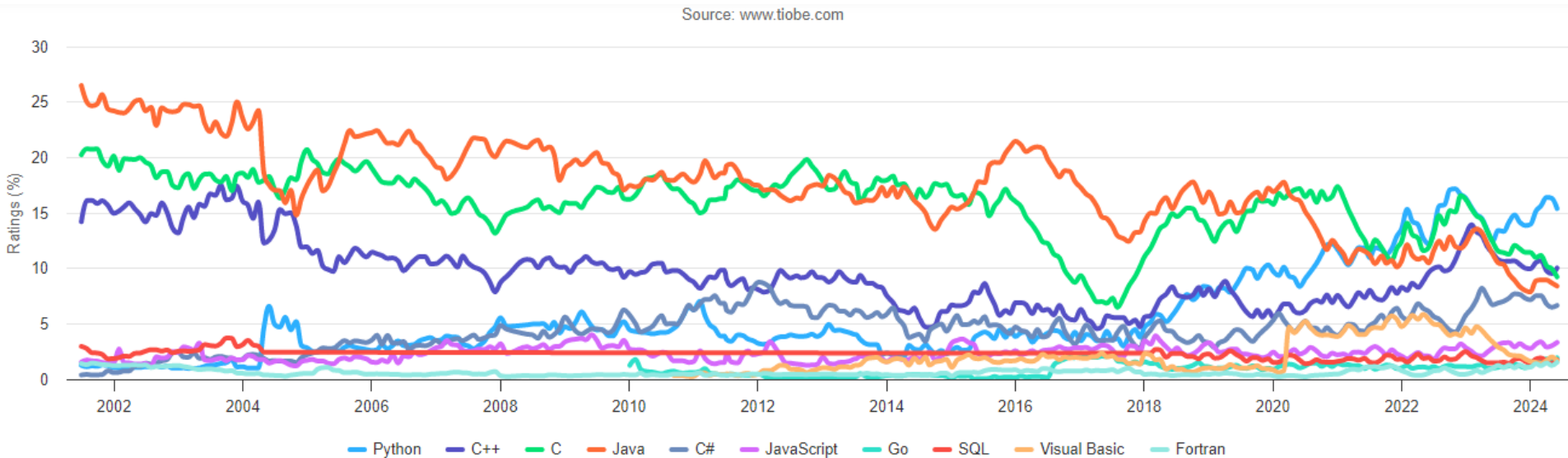


History Of Python (Timeline)



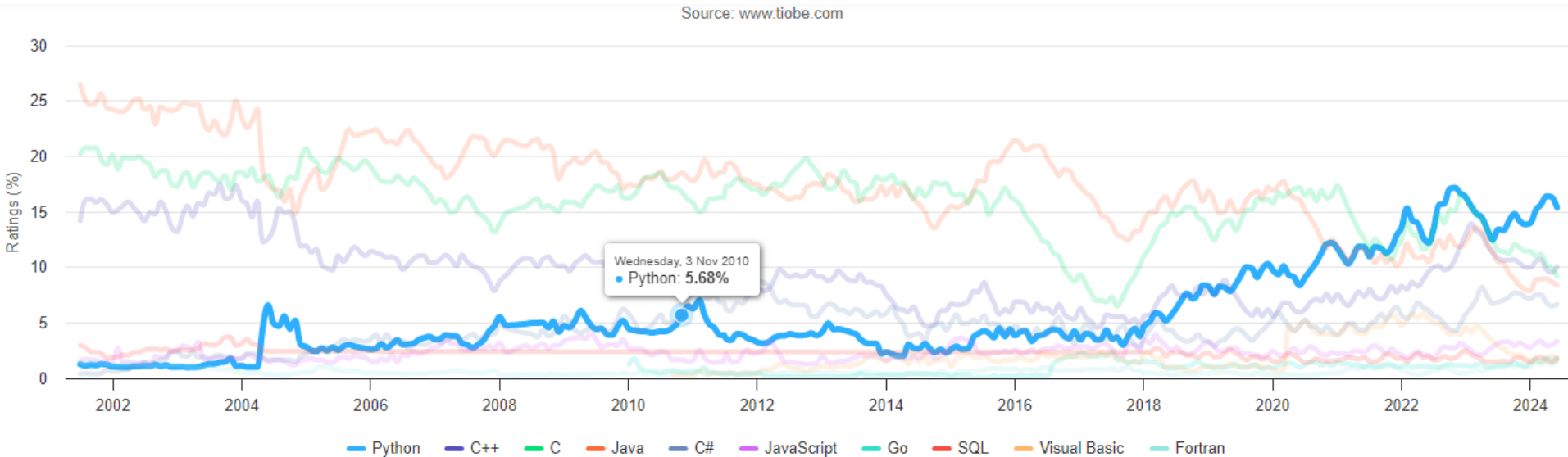
Python was created by **Guido van Rossum**

Evolution of Python



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Evolution of Python



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Why Is Python So Popular?

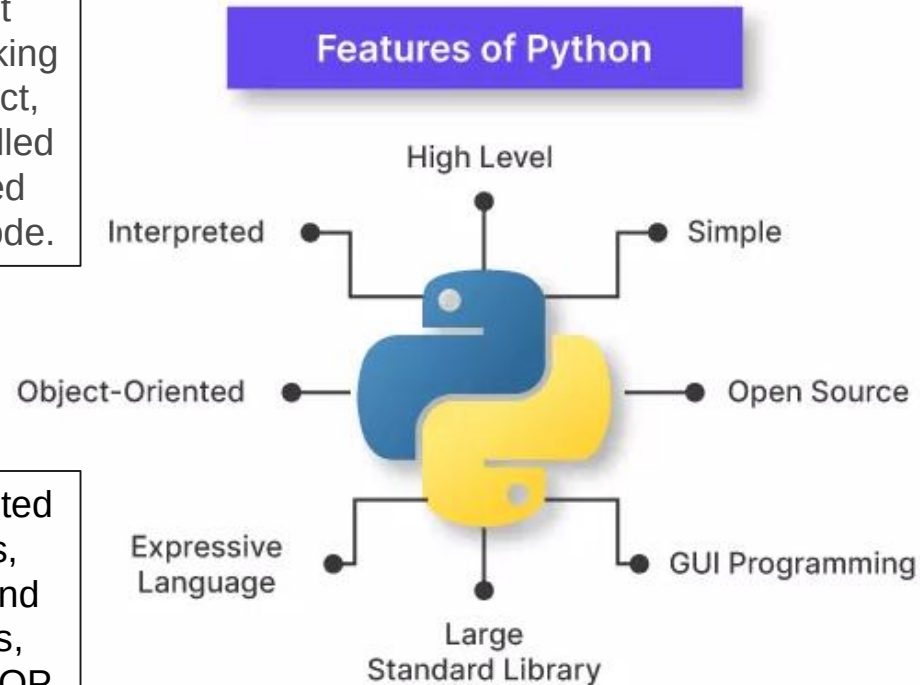


- Beginner-Friendly
- Versatility of Use
- Active and Supportive Community
- Libraries and Frameworks
- Python Works with the IoT
- Efficient and Reliable

Features of Python

Python code does not require compilation, making it easier to debug. In fact, an immediate format called the Bytecode is created from Python's source code.

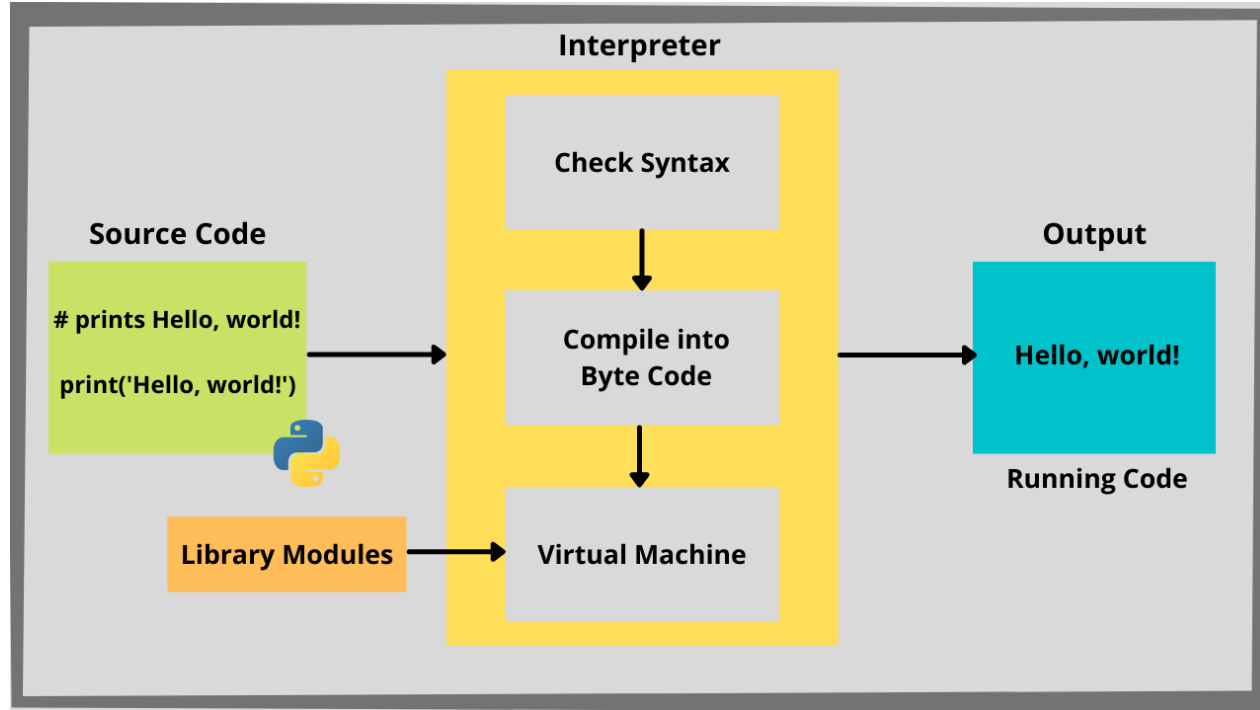
Python has object-oriented programming features, enabling the creation and use of classes, objects, inheritance and other OOP concepts



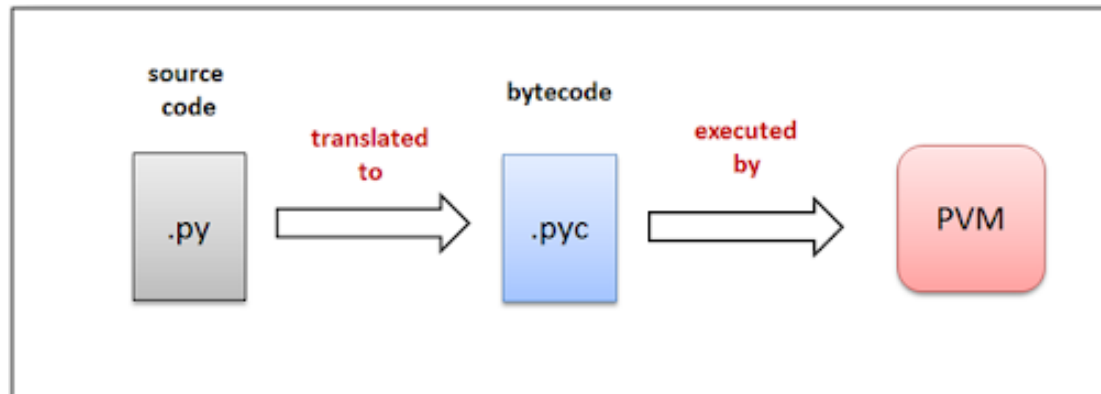
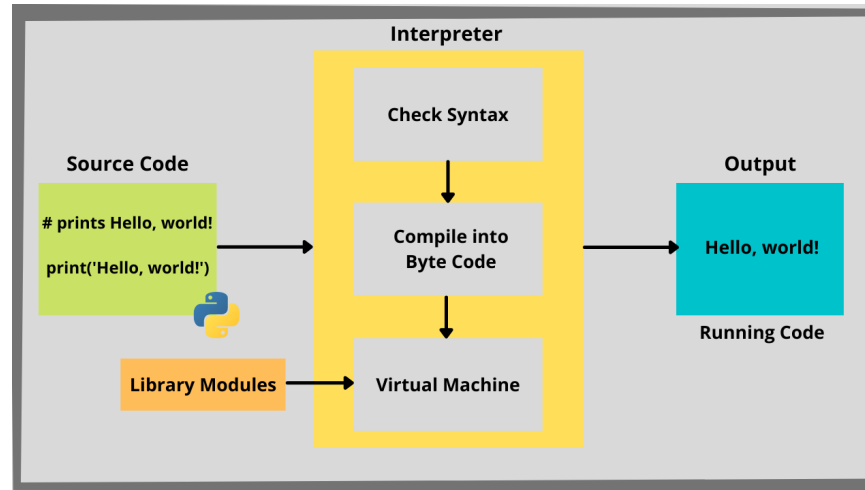
Pros and Cons of Python

Advantages			Disadvantages	
Free and Open-Source	01		Poor Memory Efficiency	
Easy to Learn	02		Slow Speed	
Vast Libraries Support	03		Database Access	
Greater Productivity	04		Weak in Mobile Computing	
Interpreted language	05		Runtime Errors	
Portability	06		Design Restrictions	
Dynamically Typed	07		Speed Limitations	

Interpreting a Python Program

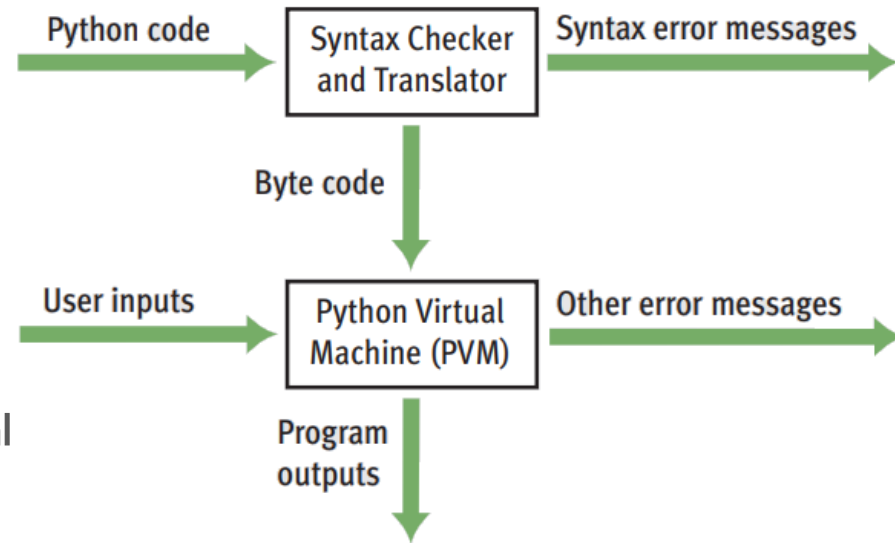


Interpreting a Python Program



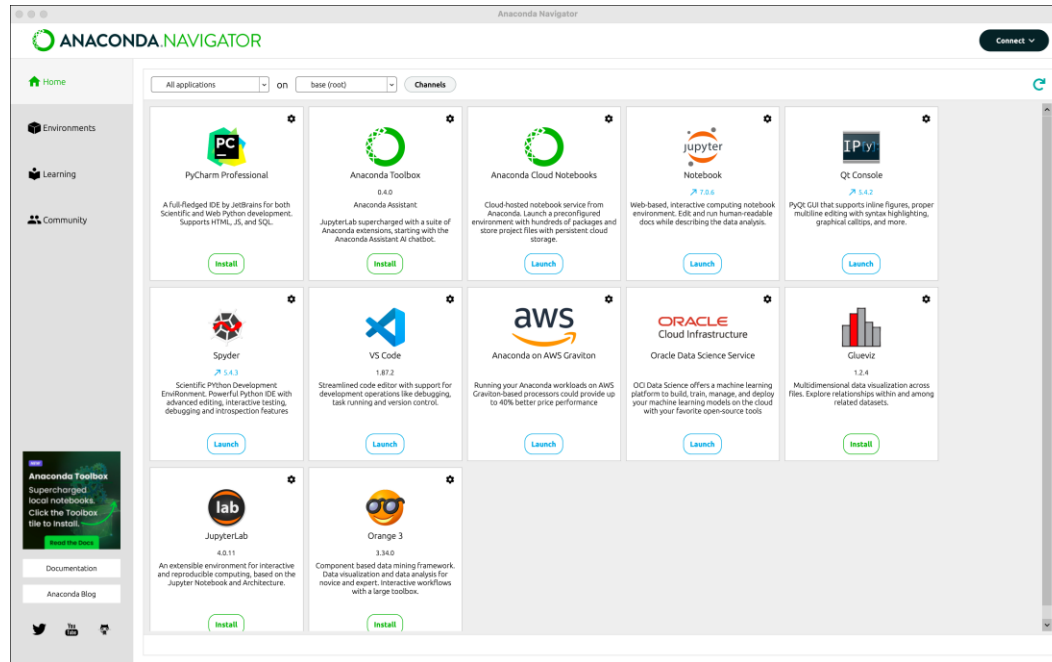
Steps in interpreting a Python program

- The interpreter reads a Python expression or statement(**source code**) and verifies that it is well formed.
- If a Python expression is well formed, the interpreter then translates it to an equivalent form in a low-level language called **byte code**. When the interpreter runs a script, it completely translates it to byte code.
- This byte code is next sent to another software component, called the **Python virtual machine (PVM)**, where it is executed.
- If another error occurs during this step, execution also halts with an error message.



Python IDE

- We will be using the Anaconda python distribution
<https://www.anaconda.com/>
- Familiarize yourself with jupyter notebook/jupyter lab



Google Classroom

The screenshot shows a web browser window with the URL <https://classroom.google.com/c/NzUzNDMyMDA5NDk0>. The page header includes the Google Classroom logo and the course name 'MCA504: Programming Logic & Tec...' by '2081 Bacth'. Below the header are tabs for 'Stream', 'Classwork', 'People', and 'Grades'. The main content area features a large banner with the course title 'MCA504: Programming Logic & Techniques using Pyt...' and the teacher's name '2081 Bacth'. To the right of the banner is a 'Customize' button. Below the banner, there is a 'Class code' section displaying '7bne3rw2' with a copy icon. To the right of the class code is an announcement box with a teacher's profile picture and the text 'Announce something to your class'. Below the announcement box is another teacher's profile picture and the name 'Roshan Tandukar'.

→ ↻ <https://classroom.google.com/c/NzUzNDMyMDA5NDk0> 🔊 🔍 ⭐ ⚙️ | ⭐ 📌

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